CS2060 Programming with C - Ch. 4 Assignment

1 Instructions

1. Submit this assignment on Canvas.
2. Solve each question in a separate (.c) file.
3. Print screen of the execution of each question (as evidence that your code is working).
4. It is highly recommended to run your code (for each question) on the test cases and print screen the results each time.
5. For each question, collect the .c file and its print screens into a folder.
6. Create a folder for the whole assignment.
7. Place all question folders into the assignment folder.
8. Compress the assignment folder into one zipped folder.
9. Upload this zipped folder to Canvas.
10. It is your own responsibility to check that you uploaded the assignment correctly (you can open the submission web page on Canvas and download your submission to check everything is OK).
11. If the Canvas is down and the deadline is too close, you can send the assignment via email to me and the TA with a print screen of the Canvas page that is down (as evidence that the Canvas website is down).

2 Questions

1. Write a C program to perform the following:
    - Asking the user to enter using the keyboard the number of lines (positive integer number).

- Printing the following star patterns separately, one below the other:

```
Please, enter the number of lines (positive integer): 0
Invalid input.
```

```
Please, enter the number of lines (positive integer): -1
Invalid input.
```

```
Please, enter the number of lines (positive integer): 5
*
**
***
****
*****

*****
****
***
**
*

*****
 ****
  ***
   **
    *

    *
   **
  ***
 ****
*****
```

2. Write a C program to perform the following:

   - Asking the user to enter using the keyboard the number of terms (positive integer).
   - Calculating the value of π from the infinite series.

$$\pi = 4 - \frac{4}{3} + \frac{4}{5} - \frac{4}{7} + \frac{4}{9} - \frac{4}{11} + ...$$

$$\pi = \frac{4}{1} - \frac{4}{3} + \frac{4}{5} - \frac{4}{7} + \frac{4}{9} - \frac{4}{11} + ...$$

```
Please, enter the number of terms (positive integer): 0
Invalid input.
```

```
Please, enter the number of terms (positive integer): -1
Invalid input.
```

```
Please, enter the number of terms (positive integer): 1000
After calculating 1000 term(s), the value of PI is 3.140593
```

3. Write a C program to perform the following:

   - Repeatedly asking the user to enter using the keyboard a number from 1 to 28 until entering -1.
   - Then, printing the corresponding character according to the following criteria:
     - the numbers correspond to the letters of the alphabet in reverse order: a = 26, z = 1, ...
     - '!' and '?' are represented by 27 and 28, respectively.

```
Please, enter a number (from 1 to 28): 1
The character is: z
Please, enter a number (from 1 to 28): 26
The character is: a
Please, enter a number (from 1 to 28): 15
The character is: l
Please, enter a number (from 1 to 28): 16
The character is: k
Please, enter a number (from 1 to 28): 17
The character is: j
Please, enter a number (from 1 to 28): 27
The character is: !
Please, enter a number (from 1 to 28): 28
The character is: ?
Please, enter a number (from 1 to 28): 0
Thank you.
```

4. Write a C program to perform the following:

   - Presenting a calculator to the user.
   - Asking the user to enter the operation (+, -, *, or /) (floating-point division is applied) (q to exit).
   - Asking the user to enter the first operand using the keyboard.
   - Asking the user to enter the second operand using the keyboard.
   - **Hint:** use the following statement to ensure that any leftover newline (or other whitespace) is skipped, and the menu is not printed twice for one operation.

```
// use scanf() with a space before %c. The space in the format
    string tells scanf to skip any whitespace characters
    (including newlines, tabs, and spaces) before reading a
    non-whitespace character.
scanf(" %c", &operation);
```

```
+ for addition.
- for subtraction.
* for multiplication.
/ for floating-point division.
q to exit.

Please, enter the operation: +
Please, enter the first operand: 1
Please, enter the second operand: 2
1.00 + 2.00 = 3.00

+ for addition.
- for subtraction.
* for multiplication.
/ for floating-point division.
q to exit.

Please, enter the operation: -
Please, enter the first operand: 1
Please, enter the second operand: 2
1.00 - 2.00 = -1.00

+ for addition.
- for subtraction.
* for multiplication.
/ for floating-point division.
q to exit.

Please, enter the operation: *
Please, enter the first operand: 1
Please, enter the second operand: 2
1.00 * 2.00 = 2.00

+ for addition.
- for subtraction.
* for multiplication.
/ for floating-point division.
q to exit.

Please, enter the operation: /
Please, enter the first operand: 1
Please, enter the second operand: 0
Division by zero is not allowed.

+ for addition.
- for subtraction.
* for multiplication.
/ for floating-point division.
q to exit.
```

```
Please, enter the operation: /
Please, enter the first operand: 1
Please, enter the second operand: 2
1.00 / 2.00 = 0.50

+ for addition.
- for subtraction.
* for multiplication.
/ for floating-point division.
q to exit.

Please, enter the operation: a
Incorrect operation.

+ for addition.
- for subtraction.
* for multiplication.
/ for floating-point division.
q to exit.

Please, enter the operation: q
Thank you.
```